Degenerate Optima in Likelihood-Ratio-Based Threshold Selection for Binary Classifiers

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Abstract

Diagnostic likelihood ratios (LR^+, LR^-) and the diagnostic odds ratio ($DOR = LR^+/LR^-$) are widely used single-number summaries of a binary classifier's usefulness at a fixed decision threshold, and are the basis of the Fagan nomogram for converting a pre-test probability into a post-test probability. A natural instinct, when a classifier's score is continuous, is to choose the operating threshold that maximizes DOR. We show this instinct is unsound: DOR is unbounded and is maximized, in general, at degenerate thresholds where sensitivity or specificity collapses toward its extreme, not at a balanced operating point. We give a short proof of this divergence along the boundary of the achievable ROC curve, extend the argument to a composite score that folds in the (threshold-invariant) area under the ROC curve, and confirm both results empirically on five-fold cross-validated ensembles from a real binary classification benchmark (n = 1,289 and n = 722). We contrast this with Youden's J statistic, whose maximizer is bounded and empirically lands at a balanced, usable operating point with comparable DOR. We conclude with a constrained variant of DOR-maximization that avoids the pathology.

1 Introduction

Let a binary classifier produce a continuous score $s(x) \in [0, 1]$ for an input x, and let $\hat{Y}_t(x) = \mathbf{1}[s(x) \geq t]$ be the decision rule at threshold t. Write sens(t) and spec(t) for the resulting sensitivity and specificity against a fixed labeled sample. The positive and negative likelihood ratios at threshold t are

$$LR^+(t) = \frac{\text{sens}(t)}{1 - \text{spec}(t)}, \qquad LR^-(t) = \frac{1 - \text{sens}(t)}{\text{spec}(t)}, \tag{1}$$

and the diagnostic odds ratio is $DOR(t) = LR^+(t)/LR^-(t)$. Equivalently, writing TP, TN, FP, FN for the threshold-t confusion-matrix counts, $DOR(t) = (TP \cdot TN)/(FP \cdot FN)$, the classical cross-product odds ratio of the 2×2 table. Likelihood ratios and DOR remain the standard tools recommended by the evidence-based-medicine literature for interpreting a diagnostic test result [Jaeschke et al., 1994a,b, Deeks and Altman, 2004, Grimes and Schulz, 2005, Altman and Bland, 1994a,b, van Stralen et al., 2009, Šimundić, 2009], and DOR in particular is a popular meta-analytic summary statistic in diagnostic-accuracy research [Glas et al., 2003] precisely because it collapses sensitivity and specificity into one number that is symmetric in the two error types and invariant to the choice of which class is labeled "positive."

That symmetry makes $DOR(t)$ tempting to use directly as a threshold-search objective: rather than picking $t = 0.5$ by convention, or minimizing misclassification rate (which is itself known to be a poor objective under class imbalance [Provost et al., 1998, He and Garcia, 2009, Chawla et al., 2002], since it is minimized by simply predicting the majority class as the imbalance grows), one might search for $t^* = \arg\max_t DOR(t)$. We show this is a mistake, with a short, general argument, and confirm the failure mode empirically. Our result formalizes, at the level of a single threshold search, a concern already raised at the level of comparing markers across studies: Pepe et al. [2004] showed that a marker can have a large odds ratio while contributing almost nothing to discrimination, i.e. that the odds ratio is a poor stand-alone gauge of diagnostic performance. We show the same statistic is not merely uninformative but actively misleading when used as a search objective over thresholds for a *single* marker.

2 Related work

ROC analysis and AUC. The receiver operating characteristic (ROC) curve and the area under it (AUC) are the standard tools for describing a classifier's discrimination independent of any single threshold [Metz, 1978, Swets, 1988, Hanley and McNeil, 1982, Zweig and Campbell, 1993, Obuchowski, 2003, Zou et al., 2007, Fawcett, 2006]. The framework originates in signal detection theory [Green and Swets, 1966], developed to characterize an observer's ability to distinguish signal from noise in radar and psychophysical tasks, well before its adoption in radiology and clinical epidemiology. Hanley and McNeil [1982] established the now-standard interpretation of AUC as the probability that a randomly chosen positive case is ranked above a randomly chosen negative case, equivalent to the Mann–Whitney U statistic, a fact we use directly in Section 5. DeLong et al. [1988] gave the widely used nonparametric method for comparing correlated AUCs, the threshold-free analogue of the threshold-specific comparisons we perform here. McClish [1989] proposed restricting AUC to a clinically relevant range of false-positive rates (partial AUC), a threshold-*range* restriction in the same spirit as, though weaker than, the threshold- *point* floor we propose in Section 7. Under class imbalance specifically, Davis and Goadrich [2006] and Saito and Rehmsmeier [2015] argue that precision-recall curves are more informative than ROC curves, since a classifier's precision (and hence positive likelihood behavior) is directly sensitive to the class balance in a way sensitivity and specificity are not; this is a complementary observation to ours, not a competing one – DOR-maximization is degenerate on *either* imbalanced or balanced data, since Theorem 1 does not depend on the class ratio, only on the existence of a boundary configuration.

Optimality and the Neyman–Pearson lemma. The likelihood-ratio statistic underlying $LR^+(t)$ and $LR^-(t)$ is not an arbitrary choice: the Neyman–Pearson lemma [Neyman and Pearson, 1933] establishes that, for testing a simple null against a simple alternative, thresholding the likelihood ratio is the *uniformly most powerful* test at any fixed significance level. This classical result explains why likelihood ratios are the natural currency for evaluating a diagnostic test result at a *given* threshold. It says nothing, however, about which threshold to prefer, since the lemma fixes the significance level (equivalently, $1 - \text{spec}(t)$) as given rather than optimized; our result can be read as showing what goes wrong when that omission is filled in naively by optimizing a ratio of two Neyman–Pearson-optimal quantities against each other, rather than by fixing an operating constraint and optimizing power subject to it, which is the correct generalization of the lemma's own logic.

Likelihood ratios, the Fagan nomogram, and diagnostic odds ratio. Fagan [1975] introduced the nomogram construction for converting a pre-test probability into a post-test probability given a likelihood ratio, later popularized for clinical use by the "Users' Guides" series [Jaeschke et al., 1994a,b] and the BMJ Statistics Notes series [Altman and Bland, 1994a,b,c, Deeks and Altman, 2004], and reviewed comprehensively by Grimes and Schulz [2005], Šimundić [2009], and van

Stralen et al. [2009]. Glas et al. [2003] and Pepe et al. [2004] both study DOR as a summary statistic – the former establishing it as a standard meta-analytic tool, the latter cautioning against its use as a stand-alone performance gauge, a concern this paper extends specifically to threshold selection. Meta-analytic treatments of sensitivity and specificity jointly, rather than via the scalar DOR, are given by Rutter and Gatsonis [2001] and Reitsma et al. [2005], whose bivariate models sidestep exactly the collapsing-to-a-single-number issue we identify here.

Threshold and cutpoint selection. Youden's index [Youden, 1950] remains the most widely used bounded criterion for selecting an "optimal" operating point from an ROC curve. Perkins and Schisterman [2006] showed that Youden's index and the point-closest-to-$(0, 1)$ criterion can select substantially different thresholds from one another, a cautionary result in the same spirit as ours but concerning two already-well-behaved criteria rather than an unbounded one. Unal [2017] and Habibzadeh et al. [2016] survey the broader landscape of cutpoint-selection criteria for continuous diagnostic tests; to our knowledge, none of these surveys treats DOR itself as a direct optimization target, which is the specific gap this paper addresses.

Beyond discrimination: calibration, decision curves, and reporting. Optimizing any single discrimination statistic, bounded or not, addresses only one facet of what makes a prediction model useful. Van Calster et al. [2019] and Gneiting and Raftery [2007] argue for calibration and proper scoring rules as complementary (and sometimes more clinically important) criteria than discrimination alone; Vickers and Elkin [2006] propose decision curve analysis as a threshold-selection alternative grounded directly in net clinical benefit rather than a ratio statistic; Steyerberg et al. [2010] give a broader framework of traditional and novel performance measures for prediction models. The TRIPOD reporting guideline [Collins et al., 2015, Moons et al., 2015] and standard texts on diagnostic test evaluation [Pepe, 2003, Kraemer, 1992, Hosmer et al., 2013] likewise treat threshold choice as a reporting decision to be justified and documented, not a free parameter to be swept in search of the largest possible ratio – the practice this paper argues against.

Ensembles and model combination. Our empirical section uses ensembles combining several base classifiers, following the general justification for ensembling given by Dietterich [2000], Breiman [1996] (bagging), and Wolpert [1992] (stacked generalization); none of these works addresses threshold selection downstream of the ensemble, which is the question this paper is concerned with. We also draw on the bootstrap [Efron and Tibshirani, 1993] for uncertainty quantification in the broader empirical pipeline this paper's benchmark is drawn from, and note that the natural-frequency framing of post-test probability advocated by Gigerenzer and Hoffrage [1995] is the same framing that makes a degenerate operating point easy to recognize once stated as "out of 1000 patients" rather than as an abstract ratio – a point we return to briefly in Section 8.

3 Preliminaries

Fix a finite labeled sample of size n with n_1 positive and $n_0 = n - n_1$ negative examples, and a score function $s(\cdot)$ that is not a constant on either class. As t ranges over $(0, 1)$, $\text{sens}(t)$ is non-increasing and $\text{spec}(t)$ is non-decreasing (both are step functions with at most n jump points, one per distinct score value). We say t is an *interior threshold* if all four confusion-matrix counts are strictly positive at t, i.e. $TP(t), TN(t), FP(t), FN(t) > 0$; this is equivalent to $0 < \text{sens}(t) < 1$ and $0 < \text{spec}(t) < 1$.

Proposition 1 (Boundary configurations)**.** *If there exists a threshold t_1 at which $FP(t_1) = 0$ and $TP(t_1) > 0$ (some positive example scores strictly above every negative example), then* $\mathrm{spec}(t_1) = 1$ *and $LR^+(t_1) = +\infty$, while $LR^-(t_1) = 1 - \mathrm{sens}(t_1)$, which is finite and, for any classifier that is not perfectly sensitive at t_1, strictly positive. Symmetrically, if there exists t_0 at which $FN(t_0) = 0$ and $TN(t_0) > 0$, then* $\mathrm{sens}(t_0) = 1$*, $LR^-(t_0) = 0$, and $LR^+(t_0) = \mathrm{sens}(t_0)/(1 - \mathrm{spec}(t_0))$ is finite and positive for any classifier not perfectly specific at t_0.*

The proof is immediate from the definitions. The condition in Proposition 1 is generic: any score function with a non-degenerate tail (some positive example outscoring every negative example, or vice versa) satisfies it, which is the common case in practice whenever the two class-conditional score distributions have unequal support.

4 Divergence of DOR at the ROC boundary

Theorem 1. *Under the conditions of Proposition 1, $DOR(t_1) = LR^+(t_1)/LR^-(t_1) = +\infty$ and, symmetrically, $DOR(t_0) = +\infty$. Consequently, whenever such t_0 or t_1 exists,* $\sup_t DOR(t) = +\infty$*, and this supremum is approached only near the boundary of the achievable ROC curve, where either* $\mathrm{sens}(t)$ *or* $\mathrm{spec}(t)$ *is driven toward its extreme value.*

Proof. Immediate from Proposition 1: $DOR(t_1) = \infty/LR^-(t_1)$ with $LR^-(t_1) \in (0, \infty)$, hence $DOR(t_1) = +\infty$; the argument at t_0 is symmetric with $DOR(t_0) = LR^+(t_0)/0 = +\infty$. Because $DOR(t)$ is finite at any interior threshold (all four counts strictly positive implies $LR^+, LR^- \in (0, \infty)$), the value $+\infty$ can only be attained at (or approached arbitrarily closely near) a boundary configuration as described. □

For a finite sample, t_0 and t_1 as defined above are themselves attainable thresholds (not merely limits), so DOR is literally maximized at one of them whenever both exist and neither is excluded by tie-breaking. The classifier's true behavior at that threshold, however, is degenerate: at t_1, sensitivity can be arbitrarily close to 0 (the classifier flags almost no positives, but every one it does flag is correct), and at t_0, specificity can be arbitrarily close to 0 (the classifier flags almost every example as positive, catching every true positive but at the cost of overwhelming false-positive volume). Neither is a useful operating point in the ordinary sense [Hosmer et al., 2013, Pepe, 2003], yet both attain the formally optimal value of the objective.

Remark 1. *This failure mode is a close relative of a more familiar one: minimizing raw misclassification rate* $1 - \mathrm{accuracy}(t)$ *on an imbalanced sample (base rate $\pi \ll 0.5$) is minimized, in the worst case, by $t \to 1$ (predict the majority class always), achieving error rate π while sensitivity collapses to 0 – the well-documented pitfall of using raw accuracy to compare classifiers under imbalance [Provost et al., 1998, He and Garcia, 2009, Chawla et al., 2002]. DOR-maximization is a different, and in one sense more insidious, version of the same problem: unlike misclassification rate, which is bounded in $[0, 1]$ and whose degenerate minimizer is at least easy to recognize (zero sensitivity, error rate exactly equal to the base rate), DOR is unbounded, so its maximizer can look like an arbitrarily strong result precisely because it is degenerate.*

Proposition 2 (Youden's J does not diverge at the ROC boundary)**.** *Let* $J(t) = \mathrm{sens}(t) + \mathrm{spec}(t) - 1$ *[Youden, 1950]. Under the conditions of Proposition 1,* $J(t_1) = \mathrm{sens}(t_1) + 1 - 1 = \mathrm{sens}(t_1) < 1$ *and* $J(t_0) = 1 + \mathrm{spec}(t_0) - 1 = \mathrm{spec}(t_0) < 1$ *(strictly, whenever the classifier is not perfect at that boundary point) – that is, J remains strictly bounded away from divergence at exactly the two configurations where Theorem 1 shows $DOR \to \infty$. More generally, since* $\mathrm{sens}(t), \mathrm{spec}(t) \in [0, 1]$

for every threshold t, $J(t) \in [-1, 1]$ everywhere, so $\sup_t J(t)$ is always finite, regardless of the score distribution.

This is the formal reason the empirical maximizer of J in Table 4 lands at a balanced interior point rather than a boundary configuration: an unbounded objective is free to be pushed arbitrarily close to a boundary in search of an arbitrarily large value; a bounded one has nothing to gain by doing so, and is maximized wherever the achievable ROC curve is genuinely farthest from the chance diagonal. The same argument applies unchanged to any objective of the form $a \cdot \mathrm{sens}(t) + b \cdot \mathrm{spec}(t)$ for constants $a, b > 0$ (of which J, with $a = b = 1$, is the symmetric case): any *additive* combination of sensitivity and specificity is bounded and therefore safe from the divergence in Theorem 1, while any objective containing $\mathrm{sens}(t)$ or $\mathrm{spec}(t)$ (or $1-$ either) in a *denominator* – as LR^+, LR^-, and hence DOR all do – is at risk of it.

5 A composite score does not fix this

One might hope to repair the objective by folding in the area under the ROC curve (AUC), e.g. via

$$M(t) = \frac{LR^+(t)}{\sqrt{LR^-(t) \cdot (1 - \mathrm{AUC})}}, \tag{2}$$

reasoning that a model with poor overall ranking ability (low AUC) should be penalized. This does not change which threshold is chosen.

Corollary 1. AUC *does not depend on t: by the Mann–Whitney U-statistic identity [Hanley and McNeil, 1982],* $\mathrm{AUC} = \Pr(s(X^+) > s(X^-))$ *for independent X^+, X^- drawn from the positive and negative class-conditional score distributions, a single number determined by the score function and the sample, not by any particular operating threshold. Hence $c := (1 - \mathrm{AUC})^{-1/2}$ is a positive constant with respect to t, and $M(t) = c \cdot LR^+(t)/\sqrt{LR^-(t)}$. Since $t \mapsto c \cdot f(t)$ has the same arg max as $t \mapsto f(t)$ for any positive constant c,*

$$\arg\max_t M(t) \;=\; \arg\max_t \frac{LR^+(t)}{\sqrt{LR^-(t)}}, \tag{3}$$

independent of the model's AUC. In particular, replacing the true AUC in Equation 2 by any other constant in $(0, 1)$ leaves the maximizing threshold unchanged.

The AUC term in Equation 2 is therefore useful only for comparing the resulting metric *value* across different models (each with its own fixed AUC) – the same cross-model role AUC ordinarily plays in model comparison [DeLong et al., 1988, Zou et al., 2007] – never for improving the threshold search within a single model. Moreover, because $\sqrt{LR^-(t)}$ is a strictly increasing function of $LR^-(t)$ wherever $LR^-(t) \geq 0$, the boundary argument of Theorem 1 applies unchanged to $LR^+(t)/\sqrt{LR^-(t)}$: it too diverges to $+\infty$ at the same boundary configurations, merely at a slower rate in $LR^-(t) \to 0$ (order $LR^-(t)^{-1/2}$ instead of $LR^-(t)^{-1}$). We verified Corollary 1 computationally by substituting AUC $= 0.5$ for each model's true AUC and recomputing $\arg\max_t M(t)$: the maximizing threshold was identical to four decimal places in every case (Section 6).

6 Empirical illustration

We illustrate the two results on a real binary classification benchmark: early prediction of a binary severity outcome from routine admission laboratory values, on two independent public datasets ($n =$

1,289, prevalence 15.8%; $n = 722$, prevalence 19.0%). For each dataset we report two classifiers: the single best-performing model from a $\sim 2{,}000$-configuration model search (gradient-boosted trees, random forests, and deep neural architectures), and a simple-average ensemble [Dietterich, 2000, Breiman, 1996] of the top diverse models by held-out AUROC. All quantities below are computed from five-fold, stratified, out-of-fold predictions (i.e. no model ever scores a row it was trained on), so the reported thresholds and ratios reflect genuine held-out generalization, not in-sample fit. The domain and clinical interpretation of this benchmark are secondary here; what matters is that it is a real classifier evaluated on real held-out data, not a synthetic construction chosen to produce the pathology.

6.1 The default threshold is not degenerate

Table 1 reports the conventional default threshold ($t = 0.5$) for reference.

Table 1: Metrics at the default threshold $t = 0.5$.

Model	Sens.	Spec.	LR^+	LR^-	DOR
n=1289, single best	0.485	0.946	8.92	0.544	16.40
n=1289, 15-ensemble	0.368	0.971	12.47	0.652	19.12
n=722, single best	0.482	0.954	10.44	0.543	19.22
n=722, 5-ensemble	0.511	0.956	11.50	0.512	22.45

6.2 Unconstrained DOR-maximization is degenerate

Table 2 reports $t^* = \arg\max_t DOR(t)$, restricted only to interior thresholds (all four confusion-matrix counts positive) to avoid the literal t_0, t_1 boundary values of Theorem 1. Even with this restriction, every maximizer is a near-degenerate operating point.

Table 2: Interior-restricted DOR-maximizing threshold.

Model	t^*	Sens.	Spec.	LR^+	LR^-	DOR
n=1289, single best	0.106	0.995	0.330	1.49	0.015	99.96
n=1289, 15-ensemble	0.765	0.113	0.999	122.33	0.888	137.75
n=722, single best	0.002	0.993	0.313	1.45	0.023	61.91
n=722, 5-ensemble	0.017	0.993	0.412	1.69	0.018	95.28

In every case, $DOR(t^*) > DOR(0.5)$, confirming that the search "succeeds" at improving the objective, and in every case sensitivity or specificity is below 0.5 (as low as 0.11), confirming that the resulting classifier is not one an operator would actually choose to deploy: two of the four maximizers flag almost every example as positive (sensitivity ≈ 0.99, specificity ≈ 0.3–0.4), and one flags almost none (sensitivity 0.11).

6.3 The composite score inherits the pathology

Table 3 reports $t^* = \arg\max_t M(t)$ for the composite score of Equation 2, using each model's true AUC. As predicted by Corollary 1, substituting AUC = 0.5 in place of the true AUC reproduced the identical t^* in all four cases.

Table 3: Composite-score-maximizing threshold ($M(t)$, Eq. 2).

Model	AUC	t^*	Sens.	Spec.	LR^+	$M(t^*)$
n=1289, single best	0.857	0.894	0.059	0.999	63.82	173.6
n=1289, 15-ensemble	0.866	0.765	0.113	0.999	122.33	354.5
n=722, single best	0.889	0.971	0.095	0.998	55.51	175.1
n=722, 5-ensemble	0.893	0.898	0.117	0.998	68.32	222.1

Every maximizer in Table 3 has sensitivity below 0.12: folding in AUC did not merely fail to fix the pathology, it moved all four models onto the specificity-extreme boundary exclusively (the t_0-type degeneracy of Proposition 1, not the t_1-type), because the $\sqrt{\cdot}$ transform in the denominator changes the relative growth rate of the two boundary terms without removing either one.

6.4 Youden's J as a bounded contrast

For comparison, Table 4 reports the threshold maximizing Youden's $J(t) = \text{sens}(t) + \text{spec}(t) - 1$ [Youden, 1950], a criterion bounded in $[-1, 1]$ whose maximizer corresponds to the point on the empirical ROC curve farthest (in vertical distance) from the chance diagonal – one of the two classical "optimal cutpoint" criteria compared by Perkins and Schisterman [2006].

Table 4: Youden's J-maximizing threshold.

Model	t^*	Sens.	Spec.	J^*	DOR
n=1289, single best	0.273	0.740	0.800	0.540	11.40
n=1289, 15-ensemble	0.157	0.799	0.766	0.565	13.01
n=722, single best	0.046	0.825	0.798	0.623	18.63
n=722, 5-ensemble	0.061	0.883	0.739	0.622	21.35

Every Youden-optimal threshold is an interior, balanced point (sensitivity and specificity both between 0.74 and 0.88), and its DOR is comparable to, or in three of four cases higher than, the default-threshold DOR in Table 1 – without the collapse in either error rate seen in Tables 2–3. As Proposition 2 shows formally, this is not a coincidence of these four models: J cannot diverge because both $\text{sens}(t)$ and $\text{spec}(t)$ enter it additively and are individually bounded in $[0, 1]$; there is no ratio whose denominator can be driven to zero.

6.5 Prevalence of the pathology across a broader model set

Tables 1–4 report four headline model/dataset combinations (the single best model and the top-diverse ensemble, per dataset). To check that the degeneracy in Table 2 is not an artifact of that particular choice of two models per dataset, we repeated the interior-restricted DOR-maximization search independently for all $15 + 15 = 30$ individual base models used to build the two ensembles – a diverse set spanning gradient-boosted trees (XGBoost, LightGBM in standard/DART/GOSS mode, CatBoost in standard/Plain mode, HistGBDT), random forests, extremely randomized trees, AdaBoost, bagging, and DNN/hybrid architectures. Every one of these 30 models was fit under the same five-fold out-of-fold protocol as the headline results. Tables 5 and 6 report the full result.

The result is unanimous: all 30 of 30 models (100%), spanning every algorithm family in the zoo, have sensitivity or specificity below 0.5 at their own interior-restricted DOR-maximizing threshold;

Table 5: Interior-restricted DOR-maximizing threshold for all 15 cached base models, multiml ($n = 1289$).

Model	AUC	t^*	Sens.	Spec.	DOR	Flag
hybrid_dnn(64,)_conv(8, 16)_gbdt100-5-0.05_gbdt_heavy	0.857	0.106	0.995	0.330	100.0	yes
hybrid_dnn(64,)_conv(16, 32)_gbdt100-5-0.05_gbdt_heavy	0.857	0.877	0.078	0.999	92.3	yes
lightgbm_n300_leaves31_lr0.05	0.854	0.993	0.064	0.999	73.8	yes
rf_n200_d15_None	0.853	0.640	0.113	0.998	68.8	yes
lightgbm_n200_leaves31_lr0.05	0.853	0.953	0.103	0.999	124.4	yes
rf_n800_d15_None	0.852	0.667	0.093	0.999	111.3	yes
extra_trees_n800_d15	0.850	0.598	0.093	0.999	111.3	yes
rf_huge_n300_leafnodes2500_None	0.850	0.657	0.103	0.999	124.4	yes
rf_huge_n300_leafnodes5000_None	0.850	0.657	0.103	0.999	124.4	yes
extra_trees_n800_d30	0.849	0.604	0.093	0.999	111.3	yes
dnn_v2_(16,)_drop0.1_lr0.005_wd0.1	0.849	0.938	0.059	0.999	67.7	yes
catboost_n800_d8_lr0.1	0.849	0.954	0.078	0.999	92.3	yes
catboost_n500_d8_lr0.1	0.848	0.926	0.088	0.999	104.9	yes
bagging_n200_ms0.8	0.848	0.720	0.103	0.999	124.4	yes
lightgbm_goss_n300_leaves63_lr0.05	0.847	0.919	0.093	0.999	111.3	yes

20 of 30 (66.7%) fall below 0.3 on whichever side is degenerate. AUC across this set ranges narrowly from 0.847 to 0.889 – these are all reasonably strong classifiers by ordinary standards – yet not one of them survives unconstrained DOR-maximization with a usable operating point. This rules out the possibility that Tables 2–3 happened to showcase two unusually pathological models: the pathology is the rule for this pathway, not the exception, across tree ensembles, boosting variants, and neural architectures alike.

7 A constrained remedy

If a ratio-based objective is preferred to J for domain reasons (e.g. LR^- has a direct clinical or decision-theoretic interpretation via the Fagan-nomogram post-test-probability calculation [Fagan, 1975] that J lacks), the pathology can be avoided by constraining the search:

$$t^*_\tau \;=\; \arg \max_{t\,:\,\min(\mathrm{sens}(t),\,\mathrm{spec}(t))\,\geq\,\tau} DOR(t), \tag{4}$$

for an explicit floor $\tau \in (0, 1)$ chosen for the application (e.g. $\tau = 0.5$). This is a strict restriction of the interior-threshold set used in Table 2, which we showed is necessary but not sufficient: interior thresholds still allow sensitivity or specificity arbitrarily close to 0. The floor τ makes the exclusion explicit and auditable, rather than implicit in whatever thresholds happen to survive the interior restriction on a given sample.

An alternative to an ad hoc floor is to make the trade-off between sens(t) and spec(t) explicit via a cost matrix, as in the cost-sensitive learning literature [Elkan, 2001, Sheng and Ling, 2006]. Elkan [2001] shows that optimal cost-sensitive classification reduces to thresholding the posterior probability at $t^* = c_{FP}/(c_{FP} + c_{FN})$, where c_{FP} and c_{FN} are the costs of a false positive and a false negative respectively; this threshold is well-defined and finite for any finite, nonzero cost pair, and by construction cannot land on the degenerate boundary configurations of Proposition 1 unless the analyst explicitly sets one cost to zero. Sheng and Ling [2006] give a practical procedure for selecting this threshold from a validation set. Compared to constrained DOR-maximization, cost-based thresholding has the advantage of making the value judgement explicit as a cost ratio

Table 6: Interior-restricted DOR-maximizing threshold for all 15 cached base models, lnn ($n = 722$).

Model	AUC	t^*	Sens.	Spec.	DOR	Flag
lightgbm_n800_leaves15_lr0.01	0.889	0.002	0.993	0.313	61.9	yes
lightgbm_n500_leaves15_lr0.01	0.889	0.921	0.102	0.998	66.5	yes
xgboost_n100_d3_lr0.2	0.889	0.006	0.993	0.450	111.1	yes
xgboost_n200_d3_lr0.2	0.887	0.001	0.993	0.378	82.6	yes
catboost_n500_d4_lr0.01	0.887	0.716	0.175	0.998	124.0	yes
histgbdt_huge_n100_leafnodes5000	0.886	0.001	0.993	0.383	84.4	yes
histgbdt_n100_lr0.1	0.886	0.001	0.993	0.383	84.4	yes
lightgbm_goss_n100_leaves31_lr0.1	0.885	0.011	0.993	0.376	82.0	yes
lightgbm_goss_n100_leaves63_lr0.1	0.885	0.011	0.993	0.376	82.0	yes
lightgbm_dart_n300_leaves15_lr0.1	0.885	0.957	0.109	0.998	71.8	yes
catboost_n800_d4_lr0.01	0.885	0.773	0.182	0.998	130.4	yes
adaboost_n300_lr0.1	0.883	0.289	0.993	0.352	73.9	yes
adaboost_n500_lr0.1	0.883	0.320	0.993	0.359	76.2	yes
catboost_plain_n100_d4_lr0.05	0.881	0.804	0.131	0.998	88.3	yes
catboost_plain_n200_d4_lr0.05	0.881	0.859	0.139	0.998	94.0	yes

rather than implicit in a floor τ, at the price of requiring that ratio to be specified in the first place – itself often difficult in a clinical setting, which is why symmetric criteria such as Youden's J remain attractive defaults.

A closely related alternative is to abandon ratio-based selection altogether in favor of decision curve analysis [Vickers and Elkin, 2006], which selects a threshold by net clinical benefit at a stated harm-to-benefit ratio – a threshold-selection rule of exactly the cost-sensitive form above, phrased in decision-analytic rather than statistical language – or to report calibration-based measures alongside discrimination [Van Calster et al., 2019, Gneiting and Raftery, 2007, Steyerberg et al., 2010] so that a degenerate operating point is visible in the reported summary rather than hidden behind a single scalar.

7.1 The degenerate maximizer is also statistically fragile

The degeneracy in Table 2 is not only a clinical-usability problem; the maximizing threshold is also a poor *statistical* estimate, because it is frequently selected by a boundary configuration resting on very few discordant cells. Table 7 reports Wilson score confidence intervals [Newcombe, 1998] for specificity at each DOR-maximizing threshold from Table 2, together with the resulting range for LR^+ obtained by propagating the interval extremes through Equation 1 – a conservative, illustrative bound rather than an exact interval for the ratio itself; a full treatment would use the sample-size and interval methods developed specifically for likelihood ratios by Simel et al. [1991].

Table 7: 95% Wilson interval for specificity at the DOR-maximizing threshold, and the resulting range for LR^+.

Model	FP / n_{neg}	Spec. (95% CI)	LR^+ point	LR^+ range
n=1289, single best	727/1085	0.330 (0.303, 0.359)	1.49	(1.43, 1.52)
n=1289, 15-ensemble	1/1085	0.999 (0.995, 1.000)	122.33	(31.4, 469.0)
n=722, single best	402/585	0.313 (0.277, 0.352)	1.44	(1.38, 1.48)
n=722, 5-ensemble	344/585	0.412 (0.373, 0.452)	1.69	(1.59, 1.75)

The n=1289 ensemble row is the most striking: its reported DOR-maximizing LR^+ of 122.33 rests on a single false positive out of 1,085 negative examples ($FP = 1$). The 95% interval for specificity at that point, $(0.995, 1.000)$, propagates to an LR^+ range of roughly $(31, 469)$ – more than an order of magnitude wide, from a single misclassified example either way. A threshold selected this close to a sample-specific accident is not a reproducible operating point; it is a sampling artifact that happens to maximize an unbounded statistic. This is a second, independent reason (beyond the low sensitivity documented in Table 2) not to trust an unconstrained DOR-maximizing threshold: it is simultaneously clinically extreme and statistically unstable.

8 Discussion

The underlying issue is general: any objective built as an unconstrained ratio of two quantities that can each independently approach 0 or ∞ along the achievable range of a one-dimensional threshold search is at risk of this failure mode, regardless of how reasonable the ratio looks at typical, non-extreme operating points. DOR is a particularly easy case to overlook because it is a well-established, symmetric, threshold-invariant-*sounding* summary statistic in the diagnostic-accuracy literature [Glas et al., 2003, Šimundić, 2009] – it is usually reported *at* a threshold chosen by other means (Youden's J [Youden, 1950], a clinically motivated cutoff, or the point closest to $(0, 1)$ on the ROC curve [Perkins and Schisterman, 2006]), not used *to select* that threshold. Using it as the search objective directly inverts that relationship and reintroduces the pathology this paper describes. This echoes, at the level of a single threshold, the broader caution Pepe et al. [2004] raised about odds ratios as performance gauges across studies: a large ratio, on its own, certifies neither good discrimination nor a usable operating point.

We also note, as a corollary of Corollary 1, a more general point: appending any threshold-invariant quantity (AUC, AUPRC, Brier score, or any other summary computed by integrating or aggregating over all thresholds) to a per-threshold objective cannot influence *which* threshold is selected, only the resulting objective's value. This is easy to get backwards when constructing an ad hoc composite score, since the inclusion of such a term can create the appearance of a more principled, multi-criterion objective while leaving the actual threshold-selection behavior completely unchanged.

We note that the natural-frequency framing recommended for communicating post-test probability [Gigerenzer and Hoffrage, 1995] – "out of 1,000 patients like this one, how many will the test correctly flag, and how many will it wrongly flag or miss?" – makes the degeneracy in Table 2 immediately legible in a way the scalar DOR does not: a threshold that flags 99.5% of patients, or one that flags only 11% of true cases, is obviously unusable once stated in those terms, even though both post formally superior DOR values. Reporting the natural-frequency breakdown alongside any ratio-based summary, as suggested by reporting guidelines for prediction models [Collins et al., 2015, Moons et al., 2015], would have made both degenerate optima in this paper immediately visible without requiring the proof given here. The stakes of getting this framing right are not hypothetical: in the classic study by Casscells et al. [1978], a large majority of physicians and medical trainees at Harvard teaching hospitals, when given a test's sensitivity, specificity, and the disease's population prevalence, substantially overestimated the post-test probability of disease given a positive result – the same base-rate neglect that makes a boundary-configuration threshold's inflated LR^+ dangerous rather than merely inconvenient: a clinician trusting the headline ratio, without the natural-frequency context, has no way to notice that the underlying sensitivity is 11%.

Finally, a search of the kind performed in Table 2 – scanning every achievable threshold in a finite sample for the one that maximizes an unbounded statistic – is itself a multiple-comparisons problem, of the same family addressed by the false discovery rate literature [Benjamini and Hochberg, 1995] for scanning many hypotheses rather than many thresholds. We did not apply a multiplicity correction here, since our purpose is to exhibit that the unconstrained supremum is degenerate at all, not to estimate a corrected p-value for it; but the same look-elsewhere logic that motivates multiplicity correction across hypotheses motivates skepticism of the single best threshold selected from a fine grid of several hundred candidates, independent of the specific divergence argument of Theorem 1.

9 Reproducibility

All reported quantities were computed from real five-fold, stratified, out-of-fold cross-validated predictions (not simulated), using the open-source implementation available at `https://github.com/netanelcyber/PenuX-AP-Severity` (`scripts/threshold_sweep.py`). Interior-threshold and Youden's-J searches, the AUC-substitution check for Corollary 1, and all table values in Section 6 are reproducible by running that script against the two public, de-identified datasets referenced in the repository.

References

Altman, D.G., Bland, J.M. (1994). Diagnostic tests 1: sensitivity and specificity. *BMJ*, 308(6943), 1552.

Altman, D.G., Bland, J.M. (1994). Diagnostic tests 2: predictive values. *BMJ*, 309(6947), 102.

Altman, D.G., Bland, J.M. (1994). Diagnostic tests 3: receiver operating characteristic plots. *BMJ*, 309(6948), 188.

Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300.

Breiman, L. (1996). Bagging predictors. *Machine Learning*, 24(2), 123–140.

Casscells, W., Schoenberger, A., Grayboys, T.B. (1978). Interpretation by physicians of clinical laboratory results. *New England Journal of Medicine*, 299(18), 999–1001.

Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.

Collins, G.S., Reitsma, J.B., Altman, D.G., Moons, K.G.M. (2015). Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *Annals of Internal Medicine*, 162(1), 55–63.

Davis, J., Goadrich, M. (2006). The relationship between Precision-Recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning*, 233–240. ACM, New York.

Deeks, J.J., Altman, D.G. (2004). Diagnostic tests 4: likelihood ratios. *BMJ*, 329(7458), 168–169.

DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3), 837–845.

Dietterich, T.G. (2000). Ensemble methods in machine learning. In *Multiple Classifier Systems (MCS 2000)*, Lecture Notes in Computer Science, Vol. 1857, 1–15. Springer, Berlin, Heidelberg.

Efron, B., Tibshirani, R.J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall, New York.

Elkan, C. (2001). The foundations of cost-sensitive learning. In *Proceedings of the 17th International Joint Conference on Artificial Intelligence*, 973–978. Morgan Kaufmann, San Francisco, CA.

Fagan, T.J. (1975). Nomogram for Bayes's theorem. *New England Journal of Medicine*, 293(5), 257.

Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874.

Gigerenzer, G., Hoffrage, U. (1995). How to improve Bayesian reasoning without instruction: frequency formats. *Psychological Review*, 102(4), 684–704.

Glas, A.S., Lijmer, J.G., Prins, M.H., Bonsel, G.J., Bossuyt, P.M.M. (2003). The diagnostic odds ratio: a single indicator of test performance. *Journal of Clinical Epidemiology*, 56(11), 1129–1135.

Gneiting, T., Raftery, A.E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378.

Green, D.M., Swets, J.A. (1966). *Signal Detection Theory and Psychophysics*. Wiley, New York.

Grimes, D.A., Schulz, K.F. (2005). Refining clinical diagnosis with likelihood ratios. *The Lancet*, 365(9469), 1500–1505.

Habibzadeh, F., Habibzadeh, P., Yadollahie, M. (2016). On determining the most appropriate test cut-off value: the case of tests with continuous results. *Biochemia Medica*, 26(3), 297–307.

Hanley, J.A., McNeil, B.J. (1982). The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1), 29–36.

He, H., Garcia, E.A. (2009). Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9), 1263–1284.

Hosmer, D.W., Lemeshow, S., Sturdivant, R.X. (2013). *Applied Logistic Regression*, 3rd edition. John Wiley & Sons, Hoboken, NJ.

Jaeschke, R., Guyatt, G., Sackett, D.L. (1994). Users' guides to the medical literature. III. How to use an article about a diagnostic test. A. Are the results of the study valid? *JAMA*, 271(5), 389–391.

Jaeschke, R., Guyatt, G.H., Sackett, D.L. (1994). Users' guides to the medical literature. III. How to use an article about a diagnostic test. B. What are the results and will they help me in caring for my patients? *JAMA*, 271(9), 703–707.

Kraemer, H.C. (1992). *Evaluating Medical Tests: Objective and Quantitative Guidelines*. Sage Publications, Newbury Park, CA.

McClish, D.K. (1989). Analyzing a portion of the ROC curve. *Medical Decision Making*, 9(3), 190–195.

Metz, C.E. (1978). Basic principles of ROC analysis. *Seminars in Nuclear Medicine*, 8(4), 283–298.

Moons, K.G.M., Altman, D.G., Reitsma, J.B., Ioannidis, J.P.A., Macaskill, P., Steyerberg, E.W., Vickers, A.J., Ransohoff, D.F., Collins, G.S. (2015). Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): explanation and elaboration. *Annals of Internal Medicine*, 162(1), W1–W73.

Neyman, J., Pearson, E.S. (1933). On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society of London. Series A*, 231, 289–337.

Newcombe, R.G. (1998). Two-sided confidence intervals for the single proportion: comparison of seven methods. *Statistics in Medicine*, 17(8), 857–872.

Obuchowski, N.A. (2003). Receiver operating characteristic curves and their use in radiology. *Radiology*, 229(1), 3–8.

Pepe, M.S. (2003). *The Statistical Evaluation of Medical Tests for Classification and Prediction.* Oxford University Press, New York.

Pepe, M.S., Janes, H., Longton, G., Leisenring, W., Newcomb, P. (2004). Limitations of the odds ratio in gauging the performance of a diagnostic, prognostic, or screening marker. *American Journal of Epidemiology*, 159(9), 882–890.

Perkins, N.J., Schisterman, E.F. (2006). The inconsistency of "optimal" cutpoints obtained using two criteria based on the receiver operating characteristic curve. *American Journal of Epidemiology*, 163(7), 670–675.

Provost, F., Fawcett, T., Kohavi, R. (1998). The case against accuracy estimation for comparing induction algorithms. In *Proceedings of the 15th International Conference on Machine Learning*, 445–453. Morgan Kaufmann, San Francisco, CA.

Reitsma, J.B., Glas, A.S., Rutjes, A.W.S., Scholten, R.J.P.M., Bossuyt, P.M., Zwinderman, A.H. (2005). Bivariate analysis of sensitivity and specificity produces informative summary measures in diagnostic reviews. *Journal of Clinical Epidemiology*, 58(10), 982–990.

Rutter, C.M., Gatsonis, C.A. (2001). A hierarchical regression approach to meta-analysis of diagnostic test accuracy evaluations. *Statistics in Medicine*, 20(19), 2865–2884.

Saito, T., Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3), e0118432.

Sheng, V.S., Ling, C.X. (2006). Thresholding for making classifiers cost-sensitive. In *Proceedings of the 21st National Conference on Artificial Intelligence (AAAI)*, 476–481.

Simel, D.L., Samsa, G.P., Matchar, D.B. (1991). Likelihood ratios with confidence: sample size estimation for diagnostic test studies. *Journal of Clinical Epidemiology*, 44(8), 763–770.

Šimundić, A.M. (2009). Measures of diagnostic accuracy: basic definitions. *EJIFCC*, 19(4), 203–211.

Steyerberg, E.W., Vickers, A.J., Cook, N.R., Gerds, T., Gonen, M., Obuchowski, N., Pencina, M.J., Kattan, M.W. (2010). Assessing the performance of prediction models: a framework for traditional and novel measures. *Epidemiology*, 21(1), 128–138.

Swets, J.A. (1988). Measuring the accuracy of diagnostic systems. *Science*, 240(4857), 1285–1293.

Unal, I. (2017). Defining an optimal cut-point value in ROC analysis: an alternative approach. *Computational and Mathematical Methods in Medicine*, 2017, 3762651.

Van Calster, B., McLernon, D.J., van Smeden, M., Wynants, L., Steyerberg, E.W. (2019). Calibration: the Achilles heel of predictive analytics. *BMC Medicine*, 17, 230.

van Stralen, K.J., Stel, V.S., Reitsma, J.B., Dekker, F.W., Zoccali, C., Jager, K.J. (2009). Diagnostic methods I: sensitivity, specificity, and other measures of accuracy. *Kidney International*, 75(12), 1257–1263.

Vickers, A.J., Elkin, E.B. (2006). Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making*, 26(6), 565–574.

Wolpert, D.H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.

Youden, W.J. (1950). Index for rating diagnostic tests. *Cancer*, 3(1), 32–35.

Zou, K.H., O'Malley, A.J., Mauri, L. (2007). Receiver-operating characteristic analysis for evaluating diagnostic tests and predictive models. *Circulation*, 115(5), 654–657.

Zweig, M.H., Campbell, G. (1993). Receiver-operating characteristic (ROC) plots: a fundamental evaluation tool in clinical medicine. *Clinical Chemistry*, 39(4), 561–577.